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A
MAJOR PROJECT PROPOSAL
ON
**ATELIERAI: AN AI-POWERED PERSONALIZED ART
LEARNING PLATFORM USING
RETRIEVAL-AUGMENTED GENERATION**

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
GPT	Generative Pre-trained Transformer
ITS	Intelligent Tutoring System
JWT	JSON Web Token
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
SQL	Structured Query Language
UI	User Interface
UX	User Experience
JSON	JavaScript Object Notation
PDF	Portable Document Format
URL	Uniform Resource Locator

1. Introduction

Artificial Intelligence (AI) has significantly transformed modern education by enabling personalized and interactive learning experiences. While AI-based tutoring systems are widely used in fields such as programming, mathematics, and language learning, art education continues to rely heavily on static resources, self-study, and instructor-led guidance.

Many aspiring artists use online tutorials, books, and structured courses such as Drawabox to learn drawing fundamentals. However, learners often struggle to find suitable resources, maintain a structured learning path, and receive personalized guidance based on their progress. Recent advancements in Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) provide opportunities to improve art education through intelligent tutoring systems. By combining AI with curated learning resources, educational content can be delivered in a more personalized and context-aware manner.

To address these challenges, this project proposes AtelierAI, an AI-powered art learning platform that provides intelligent tutoring, personalized learning paths, exercise recommendations, and progress tracking to support learners in mastering drawing fundamentals.

1.1 Background

Art education has traditionally relied on instructor led studio training, static textbooks, and self-paced online tutorials. Unlike domains such as mathematics or programming, where Intelligent Tutoring Systems (ITS) have matured over decades, drawing instruction is harder to formalize because skill assessment is visual and subjective rather than rule based.

Structured curricula like Drawabox have shown that breaking drawing fundamentals into progressive, deliberate practice exercises improves outcomes but such

platforms remain static: every learner follows the same fixed sequence regardless of their pace, weaknesses, or prior exposure. Meanwhile, general-purpose AI chatbots can answer art questions but have no curriculum awareness, no memory of a learner’s progress, and no grounding in vetted instructional material, making their advice inconsistent in depth and accuracy.

AtelierAI is positioned at the intersection of these two gaps: combining a structured, curated art curriculum with an LLM-based tutor that is grounded via Retrieval-Augmented Generation, and wrapping both in a system that tracks individual learner progress to personalize what is taught and recommended next.

1.2 Problem Statement

Learning drawing independently can be challenging due to scattered learning resources, lack of structured guidance, and limited personalized support. Although platforms such as Drawabox provide effective curriculum, learners often struggle to identify appropriate learning sequences, maintain consistency, and monitor their progress.

Furthermore, general AI chatbots are not specifically designed for art education and may provide inaccurate or unstructured responses. Existing art-learning platforms also lack personalized learning roadmaps and adaptive educational support.

Therefore, there is a need for an intelligent platform that combines structured art education resources, AI-assisted tutoring, personalized learning recommendations, and progress tracking within a single system.

1.3 Objectives

The primary aim of this project is to develop an AI-powered art learning platform that assists users in learning drawing fundamentals through intelligent tutoring, personalized learning paths, and progress tracking.:

- i. To develop a RAG-based AI art tutor using curated educational resources and create a centralized knowledge base for art-learning materials.

- ii. To generate personalized learning roadmaps based on user skill level and recommend practice exercises and learning resources.
- iii. To track user learning progress and achievements and develop a scalable web-based learning platform.

1.4 Scope of the Study

The scope of the proposed project are:

- i. AtelierAI targets aspiring artists, students, hobbyists, and self-learners seeking structured guidance in drawing fundamentals, regardless of their prior artistic experience.
- ii. The platform focuses on core drawing concepts including perspective, construction, anatomy, shading, line control, and composition, which form the foundation of artistic skill development.
- iii. The system operates as a web-based learning platform accessible through modern browsers on desktop and mobile devices, enabling flexible and self-paced learning.
- iv. The AI Art Mentor utilizes Retrieval-Augmented Generation (RAG) and curated educational resources to provide context-aware answers, learning guidance, and artistic explanations.
- v. The platform generates personalized learning roadmaps and exercise recommendations based on user skill level, learning goals, and learning progress.
- vi. Progress tracking and learning analytics are incorporated to monitor completed lessons, learning consistency, and overall artistic development.
- vii. The deployment strategy utilizes modern cloud technologies such as ReactJS, FastAPI, PostgreSQL, ChromaDB, and cloud-hosted AI services, ensuring scalability and cost-effective deployment.

2. LITERATURE REVIEW

This literature review examines existing research and practical systems related to AI assisted learning, Retrieval Augmented Generation (RAG), intelligent tutoring systems, personalized education, and digital art learning platforms. The objective is to identify current technological capabilities, existing limitations, and research gaps that AtelierAI aims to address.

2.1 Related Works

Anderson et al. (2009) [1] conducted foundational research on Intelligent Tutoring Systems (ITS) and demonstrated that personalized computer assisted instruction can significantly improve learning outcomes compared to conventional educational methods. Their work on the ACT programming tutor showed that one on one intelligent tutoring produced learning gains approximately twice those of traditional classroom instruction, establishing adaptive content delivery and learner progress monitoring as core principles of ITS design. Intelligent tutoring systems have since been widely adopted in mathematics, programming, and language education. More recently, Ma et al. (2014) [2] conducted a meta analysis of 107 studies and confirmed that ITS consistently outperform conventional instruction and textbook based learning, with an average effect size of 0.66 standard deviations. Despite these gains, the application of ITS within art education remains limited due to the subjective and skill based nature of artistic learning, leaving significant opportunities for innovation in this domain.

Brown et al. (2020) [3] introduced GPT 3, a large language model capable of performing a wide range of natural language processing tasks, including question answering, explanation generation, and conversational assistance, and demonstrated the potential of language models to support educational activities through interactive tutoring and personalized guidance. Subsequent developments have substantially extended these capabilities. Anil et al. (2023)[4] presented Gemini, a multimodal language model developed by Google DeepMind that achieves strong

performance 4 across text reasoning, code generation, and instruction following tasks. Touvron et al. (2023) [5] introduced Llama 2, an open source family of large language models that demonstrated competitive performance with proprietary models on a range of reasoning and dialogue benchmarks. Despite these advances, standalone language models may generate inaccurate or unverified information because responses are produced from learned representations rather than trusted educational sources. This limitation presents particular challenges for educational systems where factual accuracy and curriculum alignment are critical.

Lewis et al. (2020) [6] proposed Retrieval Augmented Generation (RAG), a framework that combines information retrieval techniques with large language models to improve response accuracy. Instead of relying solely on internal model knowledge, the system retrieves relevant information from external documents before generating a response. Their experiments on open domain question answering demonstrated significant improvements in factual correctness and knowledge grounding compared to closed book baselines. The architecture established a practical solution for reducing hallucinations while maintaining the flexibility of language models. Gao et al. (2023) [7] further advanced this field by proposing modular RAG architectures and surveying retrieval augmented approaches across diverse tasks, identifying that retrieval quality and knowledge base organization are the primary determinants of system performance. This directly informs AtelierAI's emphasis on curated, structured art educational content as the foundation of its RAG pipeline.

Drawabox [2] is a structured online drawing curriculum designed to teach artistic fundamentals such as line control, perspective, construction, and spatial reasoning through progressive exercises. The platform emphasizes understanding three dimensional forms rather than copying visual references and has become one of the most widely used free drawing education resources. Its systematic approach helps learners build strong foundational skills through deliberate practice and repetition. However, Drawabox functions as a static learning resource with a fixed curriculum sequence and provides no personalized guidance, no progress tracking

system, and no AI assisted tutoring, requiring learners to independently identify weaknesses and determine appropriate remediation activities.

Recent commercial tools including Guidart, ArtSensei, Kritik AI, and DrawDojo market various AI assisted features for art learners, based on publicly available product descriptions accessed in June 2026. Guidart offers AI assisted drawing feedback and guided exercises but does not provide a structured curriculum or personalized learning roadmap. ArtSensei provides a general AI chat interface for art related questions but lacks curriculum grounding, progress tracking, or exercise recommendation. Kritik AI focuses on AI facilitated peer review of student work without an integrated tutoring or curriculum component. DrawDojo offers targeted perspective and construction drills but does not connect these to a broader curriculum sequence or learner progress model. The table above summarizes the feature set of each platform relative to AtelierAI.

As the comparison demonstrates, each existing tool addresses an isolated capability rather than integrating structured curricula, RAG grounded tutoring, personalized roadmaps, and progress analytics into a unified system. This integrated gap is precisely what AtelierAI targets.

The reviewed literature demonstrates significant advancements in intelligent tutoring systems, large language models, Retrieval Augmented Generation, adaptive learning, recommendation systems, and learning analytics. However, current art education platforms continue to exhibit important limitations: the absence of personalized AI assisted learning environments, limited integration of structured art curricula with intelligent tutoring, insufficient learner progress tracking, and the lack of centralized knowledge bases supported by Retrieval Augmented Generation. Most existing solutions either provide educational content without personalization or AI assistance without grounding in trusted learning resources. The proposed AtelierAI platform aims to address these limitations by providing a comprehensive AI powered learning environment specifically designed for artistic skill development.

3. Related Theory

The theoretical framework of AtelierAI integrates concepts from Artificial Intelligence, Retrieval-Augmented Generation, Natural Language Processing, Personalized Learning, Educational Recommendation Systems, and Web-Based Learning Technologies. These theories provide the foundation for designing an intelligent art learning platform capable of delivering personalized educational guidance and structured artistic development.

3.1 Artificial Intelligence and Natural Language Processing

3.1.1 Large Language Models (LLMs)

Large Language Models (LLMs) are advanced neural networks trained on massive text corpora capable of understanding and generating human-like language. Models such as GPT, Gemini, and Llama can perform educational tasks including question answering, concept explanation, summarization, and conversational tutoring. In AtelierAI, the LLM functions as the AI Art Mentor, providing explanations of artistic concepts, answering learner questions, and supporting self-paced learning. However, because standalone language models may generate inaccurate information, they are combined with external knowledge retrieval mechanisms to improve reliability.

3.1.2 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) combines information retrieval systems with Large Language Models to generate responses grounded in external knowledge sources. Instead of relying solely on pre-trained knowledge, the system retrieves relevant educational content from a curated art-learning database before generating a response. This approach improves factual accuracy, reduces hallucinations, and ensures that explanations are based on trusted resources. Within AtelierAI, RAG serves as the core technology behind the AI tutoring system, en-

abling learners to receive context-aware guidance derived from drawing tutorials, anatomy references, perspective guides, and educational materials.

3.2 Information Retrieval Theory

3.2.1 Text Embeddings and Semantic Search

Text embeddings are numerical vector representations of textual information that capture semantic meaning. Similar concepts produce similar vector representations, allowing systems to retrieve information based on meaning rather than exact keyword matching. For example, queries such as “How do I draw boxes in perspective?” and “How can I construct cubes in perspective?” can retrieve related educational content despite different wording. This capability forms the basis of semantic search within AtelierAI.

3.2.2 Vector Databases

Vector databases are specialized systems designed to store and search high-dimensional embeddings efficiently. Unlike traditional databases that rely on exact matching, vector databases identify content based on similarity relationships between vectors. Technologies such as ChromaDB and FAISS enable rapid retrieval of relevant educational resources from large collections of art-learning materials. In AtelierAI, the vector database acts as the knowledge repository that supports the Retrieval-Augmented Generation pipeline.

3.3 Personalized Learning Theory

3.3.1 Adaptive Learning Systems

Adaptive learning systems modify educational content according to learner needs, skill levels, and performance. Research has shown that personalized learning improves engagement, retention, and learning efficiency by providing educational experiences tailored to individual users. AtelierAI applies adaptive learning principles by generating personalized learning roadmaps that guide learners through drawing fundamentals according to their goals and progress.

3.3.2 Educational Recommendation Systems

Recommendation systems analyze user behavior, interests, and progress to suggest relevant content and activities. In educational environments, recommendation engines help learners discover suitable resources, exercises, and revision materials while reducing information overload. AtelierAI utilizes recommendation techniques to suggest practice exercises, learning modules, and supplementary resources that support artistic improvement and long-term skill development.

3.4 Learning Analytics Theory

Learning Analytics involves collecting and analyzing educational data to monitor learner performance and improve learning outcomes. Modern educational platforms use analytics to track progress, identify strengths and weaknesses, and provide personalized recommendations. AtelierAI incorporates learning analytics to measure learning consistency, completed lessons, and artistic development, enabling learners to visualize their growth over time and remain motivated throughout the learning process.

3.5 Knowledge Management Theory

Knowledge Management Systems (KMS) are designed to organize, store, retrieve, and distribute information efficiently. Educational knowledge management focuses on maintaining structured repositories of learning materials that can be easily accessed when needed. The AtelierAI knowledge base functions as a specialized educational repository containing drawing lessons, tutorials, anatomy references, perspective guides, and practice exercises. The integration of knowledge management principles with Retrieval-Augmented Generation enables efficient and reliable educational support.

3.6 System Architecture Theory

3.6.1 Web-Based Learning Architecture

Web-based learning systems provide educational services through internet-enabled platforms accessible from multiple devices. Their advantages include accessibility, scalability, ease of maintenance, and cross-platform compatibility. AtelierAI

adopts a web-based architecture to ensure learners can access educational resources, personalized roadmaps, and AI tutoring services from both desktop and mobile devices.

3.6.2 Three-Layer System Architecture

The proposed system follows a three-layer architecture consisting of Presentation, Application, and Data layers. The Presentation Layer manages user interaction through dashboards, learning interfaces, and chat systems. The Application Layer handles business logic including authentication, roadmap generation, recommendation processing, and RAG orchestration. The Data Layer manages user information, learning progress, educational resources, and vector embeddings. This layered architecture improves maintainability, scalability, and system organization.

3.7 Conceptual Framework

The conceptual workflow of AtelierAI begins when a learner creates a profile and specifies artistic goals and skill level. The system then generates a personalized learning roadmap and recommends suitable educational resources. Users interact with the AI Art Mentor through natural language queries, which are processed through the Retrieval-Augmented Generation pipeline. Relevant educational content is retrieved from the knowledge base and supplied to the language model to generate grounded responses. User activities and completed lessons are recorded, allowing the recommendation engine and learning analytics modules to continuously adapt educational guidance according to learner progress. This integration of Artificial Intelligence, Retrieval-Augmented Generation, Personalized Learning, Recommendation Systems, and Learning Analytics forms the theoretical foundation of AtelierAI.

4. Proposed Methodology

This chapter presents the methodology adopted for the development of AtelierAI. The proposed system integrates Artificial Intelligence, Retrieval-Augmented Generation (RAG), Recommendation Systems, and Personalized Learning techniques to provide an intelligent learning environment for art education.

The methodology covers system requirements, architecture design, data collection strategies, module descriptions, database design, and implementation approaches.

4.1 Feasibility Study

A feasibility study was conducted to determine the practicality and viability of developing AtelierAI.

4.1.1 Technical Feasibility

The development of AtelierAI is technically feasible because the required technologies are readily available and widely used in modern software development. The proposed system utilizes Python and FastAPI for backend development, ReactJS for the frontend, PostgreSQL for relational data management, ChromaDB for vector storage, Sentence Transformers for embedding generation, and Gemini API for AI-powered educational assistance. These technologies are well documented, actively maintained, and supported by large developer communities. Furthermore, most of the tools are open-source or available through educational and free-tier services, making implementation achievable within the scope of the project.

4.1.2 Operational Feasibility

AtelierAI is designed for aspiring artists, students, hobbyists, and self-learners who wish to improve their drawing skills through structured guidance and personalized learning. The platform operates entirely through a web browser, requiring only internet access and basic computer literacy from users. No specialized hardware or software installation is necessary. By providing centralized learning resources, AI-assisted tutoring, personalized roadmaps, and progress tracking, the system

simplifies the learning process and enhances accessibility, making it operationally feasible for a broad range of users.

4.1.3 Social Feasibility

The growing popularity of online learning platforms and digital educational resources demonstrates a strong acceptance of technology-assisted learning among students and self-learners. Many artists already depend on online tutorials, educational videos, and learning communities to develop their skills. AtelierAI builds upon these existing learning habits by offering a centralized platform that combines structured art education with intelligent educational support. By making quality art education more accessible and personalized, the system is expected to have a positive social impact and be readily accepted by its target audience.

4.1.4 Economic Feasibility

The project is economically feasible because it can be developed using freely available technologies and educational resources. The system relies primarily on open-source software frameworks, publicly available art education materials, and free-tier cloud hosting services. The use of existing AI APIs and machine learning libraries significantly reduces development and infrastructure costs. As a result, the overall financial investment required for implementation remains relatively low while still delivering substantial educational value, making the project suitable for academic development and future expansion. While development costs are minimized through open-source frameworks and free-tier services, the team acknowledges that production deployment carries ongoing operational costs specifically Gemini API usage beyond free-tier quotas (billed per token) and cloud hosting for PostgreSQL/ChromaDB at scale. For the academic prototype, usage is expected to remain within free-tier limits (Gemini API free tier, and free-tier hosting via services such as Render or Railway). A future production deployment would require a cost model based on projected query volume per user, which is outside the scope of this academic project but is identified as a direction for future work.

4.2 Requirement Analysis

Requirements engineering was conducted to define the functional behaviors and quality attributes of AtelierAI. The requirements focus on providing personalized art education through AI tutoring, structured learning pathways, educational resource management, and learner progress monitoring.

4.2.1 Functional Requirements

R.1: AI Art Mentor

Description: The system shall provide an AI-powered educational assistant capable of answering drawing-related questions using Retrieval-Augmented Generation (RAG). The assistant shall retrieve relevant information from curated art learning resources and generate educational responses tailored to the learner's needs.

i. **R.1.1: Question Processing and Validation**

Input: User-submitted drawing or art-related question.

Output: Validated query forwarded to the retrieval pipeline.

Processing: The system validates the query, removes irrelevant content, and prepares it for semantic retrieval.

ii. **R.1.2: Educational Content Retrieval**

Input: Processed user query.

Output: Relevant educational content retrieved from the knowledge base.

Processing: The system converts the query into embeddings and performs similarity search within the vector database to identify relevant learning materials.

iii. **R.1.3: AI Response Generation**

Input: User query and retrieved educational content.

Output: Context-aware educational response.

Processing: The language model generates explanations, guidance, and learning recommendations grounded in the retrieved resources.

R.2: Art Knowledge Base Management

Description: The system shall maintain a centralized repository of educational

materials including drawing lessons, anatomy references, perspective guides, exercises, and tutorials.

i. **R.2.1: Resource Upload and Processing**

Input: Educational resources in PDF or text format.

Output: Processed and indexed educational content.

Processing: The system extracts text, cleans content, segments information into chunks, and generates embeddings for retrieval.

ii. **R.2.2: Resource Storage and Organization**

Input: Processed educational content.

Output: Structured learning repository.

Processing: Educational resources are categorized and stored for efficient retrieval and management.

R.3: Personalized Learning Roadmap

Description: The system shall generate customized learning pathways based on user skill level, artistic goals, and learning progress.

i. **R.3.1: Learner Profile Assessment**

Input: User skill level, interests, and learning objectives.

Output: Learner profile.

Processing: The system analyzes user information to determine an appropriate learning starting point.

ii. **R.3.2: Roadmap Generation**

Input: Learner profile and curriculum data.

Output: Personalized learning roadmap.

Processing: The system generates a structured sequence of lessons and exercises tailored to the learner.

R.4: Exercise Recommendation System

Description: The system shall recommend learning activities and exercises based on user progress and identified weaknesses.

i. **R.4.1: Progress Analysis**

Input: Learning history and completed lessons.

Output: Learning progress assessment.

Processing: The system evaluates learner performance and identifies areas requiring additional practice.

ii. **R.4.2: Exercise Recommendation**

Input: Progress assessment and learner profile.

Output: Recommended exercises and learning materials.

Processing: The recommendation engine suggests activities appropriate to the learner's current stage.

R.5: Progress Tracking and Analytics

Description: The system shall monitor learner activities and provide measurable indicators of artistic development.

i. **R.5.1: Progress Recording**

Input: Completed lessons and exercises.

Output: Updated learner progress data.

Processing: The system records user activities and updates completion statistics.

ii. **R.5.2: Learning Analytics Dashboard**

Input: User learning history.

Output: Progress reports and visual analytics.

Processing: The system generates dashboards displaying learning consistency, completed modules, and roadmap progress.

R.6: User Management

Description: The system shall provide secure account management and authentication services.

i. **R.6.1: Registration and Login**

Input: User credentials.

Output: Authenticated user session.

Processing: The system validates user credentials and grants access to platform resources.

R.7: Admin Management

Description: The system shall optionally provide an administrative interface for managing users, curating knowledge base content, and viewing aggregate system analytics.

- i. **R.7.1: Admin Authentication** Input: Admin credentials.
Output: Authenticated admin session with elevated privileges.
- ii. **R.7.2: Content Curation**
Input: New educational resource.
Output: Approved/indexed resource in knowledge base.
- iii. **R.7.3: System Analytics View** Input: Aggregate usage data.
Output: Dashboard of platform-wide learning metrics.

4.2.2 Non-Functional Requirements

The non-functional requirements define the performance, security, usability, and scalability standards of the proposed system.

- i. **R.8.1:** AI response generation and educational content retrieval shall occur within acceptable response times under normal operating conditions.
- ii. **R.8.2:** User authentication shall be implemented using secure JSON Web Tokens (JWT) with appropriate session management mechanisms.
- iii. **R.8.3:** All communication between the client and server shall occur through secure HTTPS protocols to ensure data privacy and integrity..
- iv. **R.8.4:** The platform shall provide a responsive user interface accessible through modern desktop and mobile web browsers.
- v. **R.8.5:** The system shall support concurrent access by multiple users without significant degradation in performance.
- vi. **R.8.6:** Educational resources and learner data shall be stored securely with appropriate backup and recovery mechanisms.

- vii. **R.8.7:** The architecture shall support future expansion including AI-powered sketch critique, computer vision integration, and mobile application deployment.

4.3 Data Collection

The effectiveness of AtelierAI depends on the quality, diversity, and organization of educational resources available within its knowledge base. Since the platform utilizes Retrieval-Augmented Generation (RAG), Personalized Learning, and Recommendation Systems, multiple forms of educational data must be collected and processed. The collected data will support knowledge retrieval, roadmap generation, exercise recommendation, and learner progress analysis.

Table 4.1: Data Sources and Collection Strategy

Data Type	Collection Method	Estimated Volume	Purpose
Drawing Fundamentals and Resources	Drawing books, tutorials, educational notes	50–100 documents	Core educational knowledge base
Perspective Drawing Materials	Drawabox lessons, perspective guides, textbooks	20–30 resources	Perspective learning support
Figure Drawing Resources	Anatomy books, figure drawing tutorials	30–50 resources	Human figure and anatomy education
Practice Exercises	Structured drawing exercises and assignments	200+ exercises	Exercise recommendation system
Learning Curriculum Data	Instructor-designed learning pathways	10–20 learning modules	Personalized roadmap generation
User Learning Progress Data	User interactions and completed lessons	Dynamic	Progress tracking and analytics

4.4 Proposed System Design

This section presents the architectural blueprint and functional decomposition of the AtelierAI platform. The proposed system integrates Retrieval-Augmented Generation (RAG), Artificial Intelligence, Personalized Learning, Recommendation Systems, and Learning Analytics to provide an intelligent and adaptive art education environment.

The system is designed to ensure scalability, maintainability, and efficient educational content delivery while supporting personalized learning experiences for users with different artistic goals and skill levels.

4.4.1 System Architecture

The proposed system adopts a multi-layer architecture comprising the Presentation Layer, Application Layer, Database Layer, Vector Database Layer, and Artificial Intelligence Layer.

- i. **Presentation Layer:** The Presentation Layer serves as the user-facing component of AtelierAI and is responsible for facilitating interaction between learners and the platform. It provides functionalities such as user registration, login, personalized learning roadmap visualization, AI Art Mentor interaction, exercise recommendations, and progress tracking dashboards. The layer is designed to offer a responsive and user-friendly learning experience accessible through both desktop and mobile devices. ReactJS and Tailwind CSS are used to develop the frontend interface due to their flexibility, performance, and modern user interface capabilities.
- ii. **Application Layer:** The Application Layer contains the core business logic of the system and acts as the intermediary between the frontend, databases, and Artificial Intelligence services. It handles user authentication, educational resource management, recommendation generation, roadmap creation, progress tracking, and Retrieval-Augmented Generation orchestration. This layer processes user requests and coordinates communication among all system components to ensure smooth platform operation. FastAPI is utilized for backend development because of its high performance, scalability, and

seamless integration with Python-based AI technologies.

- iii. **Database Layer:** The Database Layer is responsible for storing and managing structured data generated within the platform. It maintains user profiles, learning history, educational resource metadata, recommendation records, and progress analytics information. By providing efficient data storage and retrieval mechanisms, this layer ensures consistency and reliability of educational and user-related information. PostgreSQL is selected as the primary relational database due to its robustness, scalability, and support for complex data management requirements.
- iv. **Vector Database Layer:** The Vector Database Layer manages vector embeddings generated from educational resources and user queries. These embeddings enable semantic search capabilities, allowing the system to retrieve relevant learning materials based on meaning rather than exact keyword matches. The layer supports efficient similarity search operations and serves as the foundation of the RAG pipeline. ChromaDB is utilized as the vector database because of its lightweight architecture, ease of integration, and effectiveness in handling embedding-based retrieval tasks.
- v. **Artificial Intelligence Layer:** The AI Layer serves as the core intelligence component of AtelierAI. It combines embeddings, vector retrieval, and LLMs to provide personalized educational support. When a learner submits a query, the system retrieves relevant educational content from the knowledge base and combines it with the user’s question to generate context-aware responses. This layer also supports roadmap assistance, educational recommendations, and adaptive learning guidance. The primary technologies employed include Sentence Transformers for embedding generation and the Gemini API for response generation within the RAG framework.

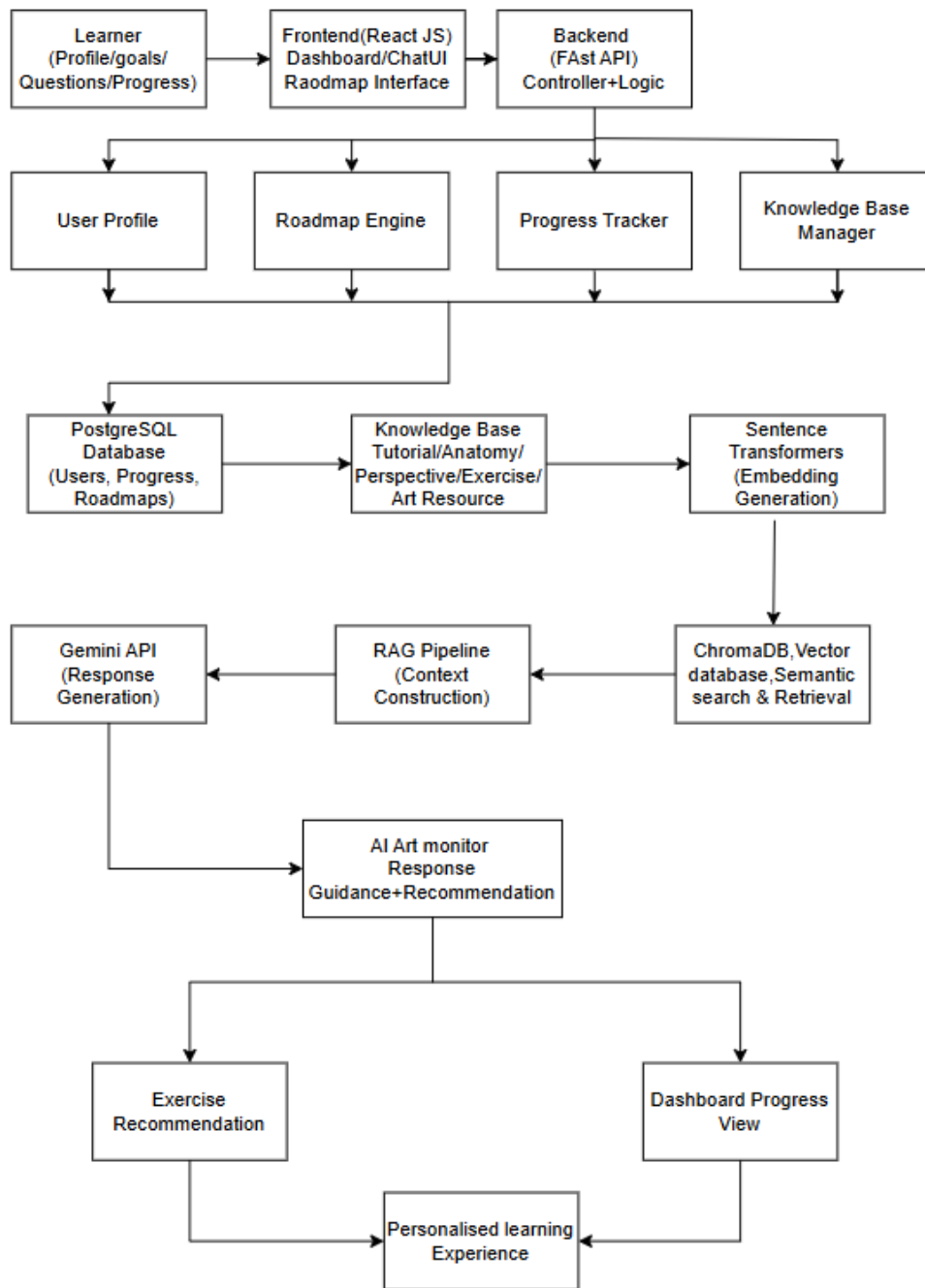


Figure 4.1: System Architecture And Block Diagram of Atelier AI

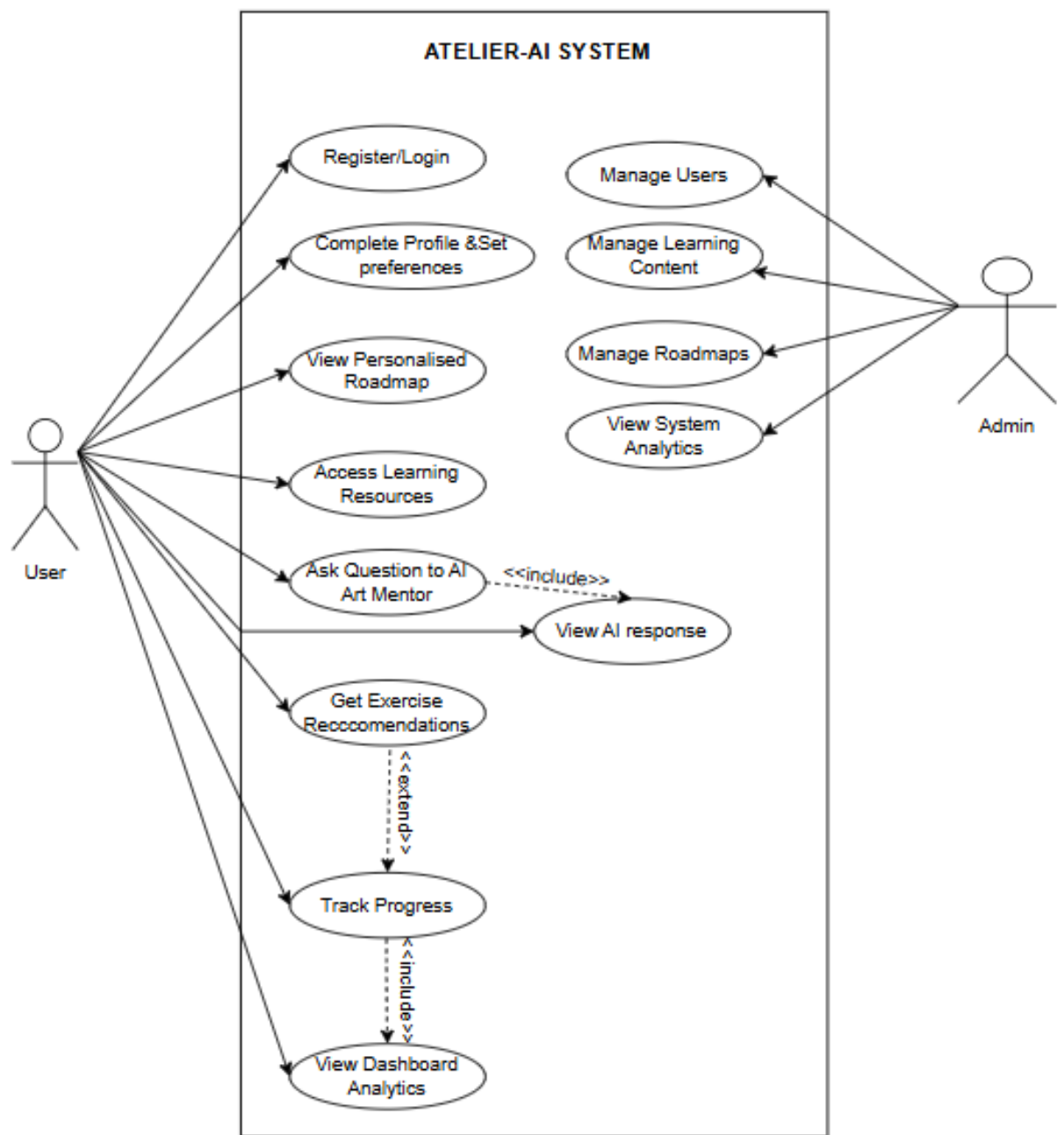


Figure 4.2: Use Case Diagram for AtelierAI

4.4.2 Technology Stack Justification

The proposed technology stack for AtelierAI consists of modern, reliable, and widely adopted technologies that support the development of intelligent educational systems. The selected technologies provide scalability, maintainability, efficient AI integration, and cost-effective deployment. Furthermore, most of the chosen tools are open-source or available through educational and free-tier services, making them suitable for academic project development.

Table 4.2: Technology Stack Components

Component	Technology	Rationale
Frontend	React.js PWA	Progressive Web App, Component-based architecture, responsive user interface, and efficient state management
Styling Framework	Tailwind CSS	Rapid UI development and responsive design support
Backend	Python FastAPI	High performance, asynchronous processing, and seamless AI integration
Relational Database	PostgreSQL	Reliable structured data management and scalability,
Vector Database	ChromaDB	Efficient storage and retrieval of vector embeddings.
Embedding Model	Sentence Transformers	High-quality semantic embeddings for educational content retrieval
Large Language Model	Gemini API	Context-aware response generation and educational assistance
Authentication	JWT	Secure user authentication and session management

4.4.3 Working Flow of System

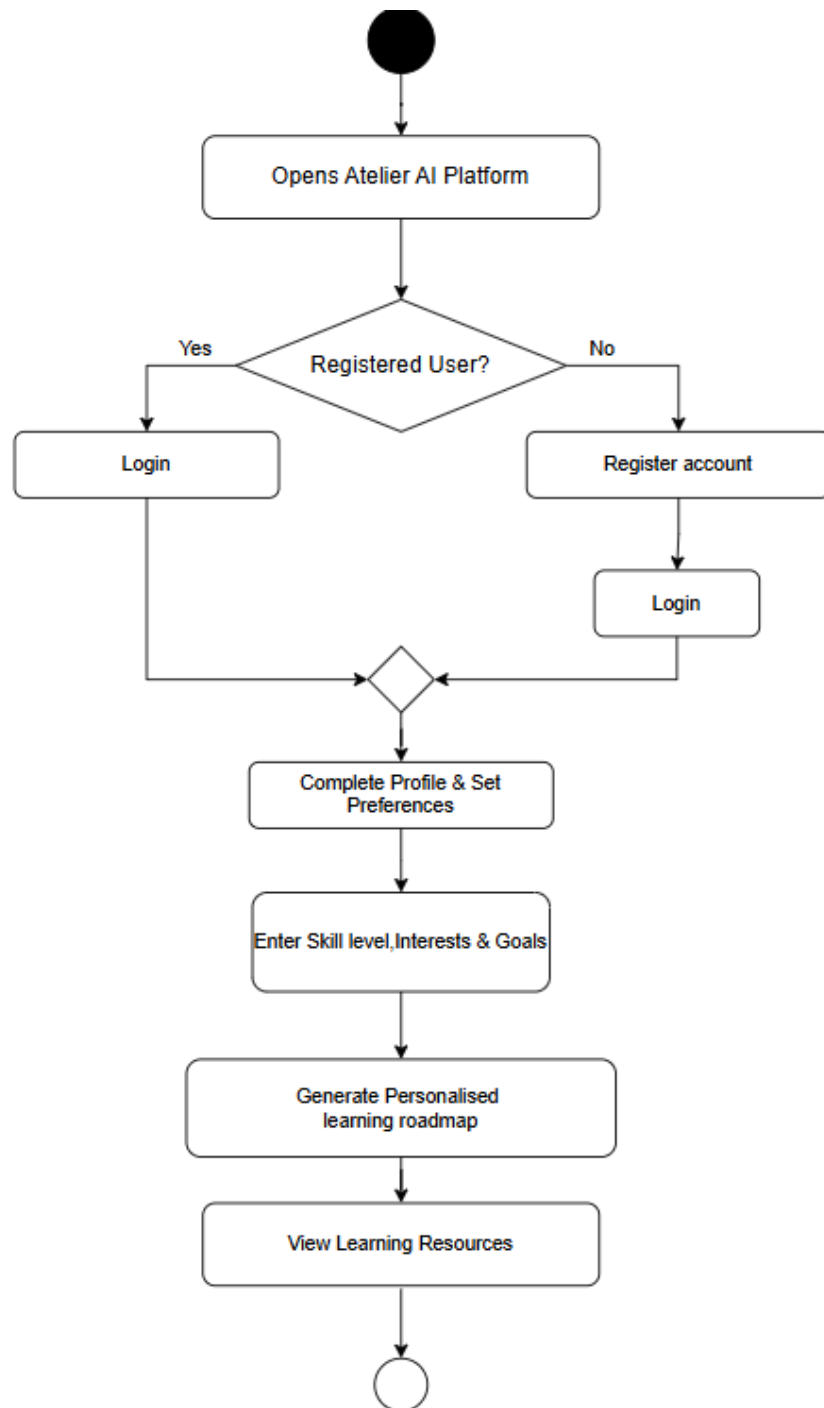


Figure 4.3: Workflow Diagram of AtelierAI (1)

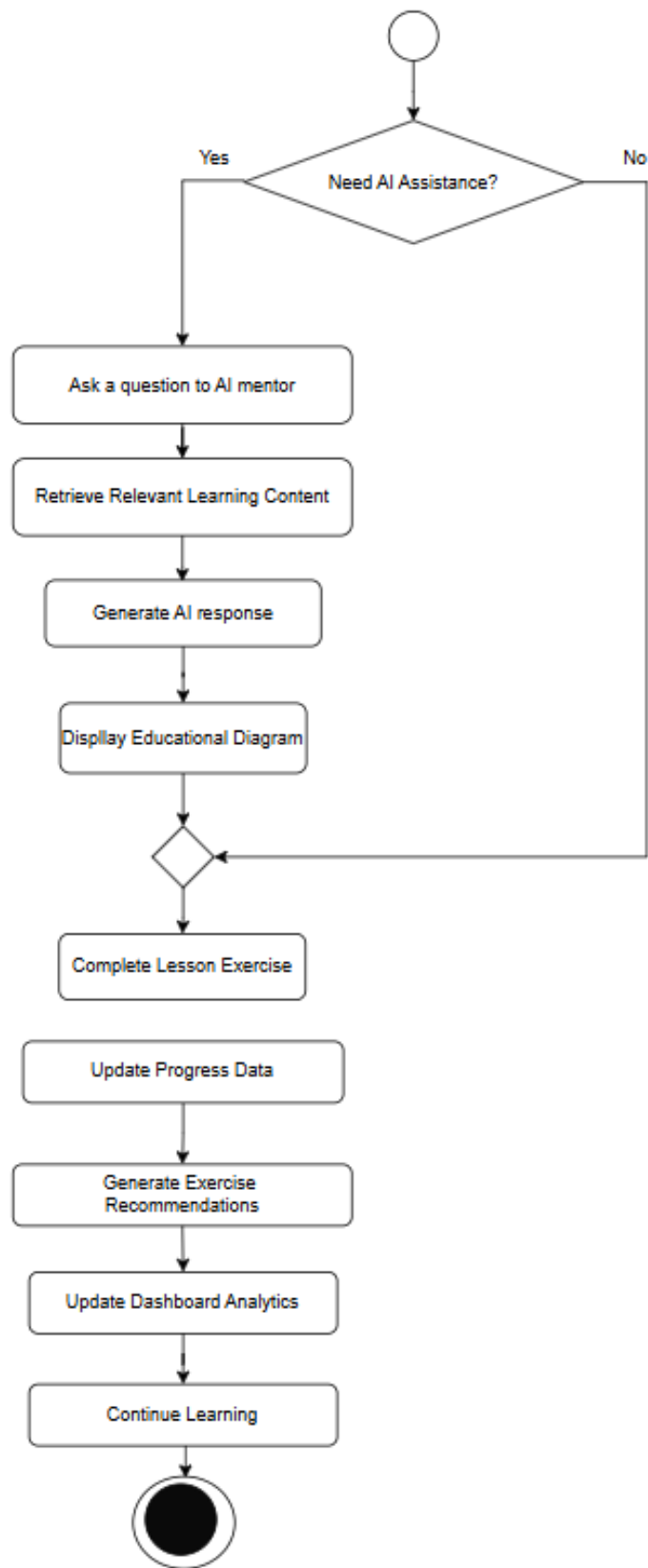


Figure 4.4: Workflow Diagram of AtelierAI(2)

The operational workflow of AtelierAI is designed to provide learners with a personalized and intelligent learning experience through personalized roadmaps, educational resources, Retrieval-Augmented Generation (RAG), and learning analytics. The workflow of the proposed system is as follows:

- i **Registration and Profile Creation:** The learner creates an account and provides information such as skill level, artistic interests, and learning objectives. The system uses this information to establish a learner profile.
- ii **Roadmap Generation:** Based on the learner profile, the system generates a personalized learning roadmap consisting of lessons, exercises, and educational resources appropriate to the learner’s goals and current skill level.
- iii **Learning Resource Access:** The learner accesses recommended educational materials, including drawing tutorials, perspective guides, anatomy references, and practice exercises through the platform.
- iv **AI Art Mentor Interaction:** The learner submits drawing-related questions through the AI Art Mentor interface using natural language.
- v **Query Processing and Retrieval:** The system converts the query into vector embeddings and performs semantic search within the vector database to retrieve relevant educational content from the knowledge base.
- vi **Response Generation:** The retrieved content is combined with the learner’s query through the Retrieval-Augmented Generation pipeline and supplied to the Gemini language model to generate context-aware educational responses and recommendations.
- vii **Progress Tracking and Recommendations:** User activities such as completed lessons, AI interactions, and exercise history are recorded. Based on this information, the system updates learner progress and recommends suitable exercises and learning resources.
- viii **Dashboard Update:** The learner dashboard is updated with roadmap progress, completed modules, analytics, and recommended next learning activities, enabling continuous and adaptive learning.

4.4.4 Module Descriptions

- i. **AI Art Mentor Module:** The AI Art Mentor Module serves as the primary educational assistance component of AtelierAI. It enables learners to ask drawing-related questions and receive intelligent, context-aware guidance through natural language interaction. The module utilizes the Retrieval-Augmented Generation (RAG) framework to retrieve relevant educational content from the knowledge base and combine it with the learner’s query before generating responses. By grounding responses in curated educational resources, the module provides accurate explanations, drawing techniques, learning recommendations, and personalized support throughout the learning process.
- ii. **Art Knowledge Base Management Module:** The Art Knowledge Base Management Module is responsible for maintaining the educational repository that supports the AI Art Mentor. It manages drawing tutorials, anatomy references, perspective guides, practice exercises, and other learning materials collected from trusted educational sources. The module performs content preprocessing, text segmentation, embedding generation, and indexing to ensure that educational resources can be efficiently retrieved through semantic search. This repository forms the foundation of the system’s knowledge retrieval process.
- iii. **Personalized Learning Roadmap Module:** The Personalized Learning Roadmap Module generates customized learning pathways based on the learner’s skill level, artistic interests, and educational objectives. By analyzing learner profiles, the module identifies an appropriate starting point and organizes lessons and exercises into a structured progression. The generated roadmap provides learners with clear guidance on what topics to study, helping them develop artistic skills systematically and efficiently.
- iv. **Exercise Recommendation Module:** The Exercise Recommendation Module assists learners by recommending suitable practice activities and learning resources according to their progress and learning needs. The module evaluates completed lessons, roadmap progression, and learning history to identify

areas that require additional practice. Based on this analysis, it suggests relevant exercises, revision materials, and supplementary educational content that support continuous improvement and reinforce learning outcomes.

- v. **Progress Tracking and Analytics Module:** The Progress Tracking and Analytics Module monitors learner activities and provides measurable indicators of artistic development. It records completed lessons, exercise participation, AI interactions, and roadmap advancement, allowing the system to evaluate learning consistency and progress over time. The collected information is presented through dashboards and analytics tools that help learners visualize their achievements, identify strengths and weaknesses, and maintain motivation throughout their learning journey.
- vi. **User Management Module:** The User Management Module handles user authentication, profile management, and access control within the platform. It enables learners to create accounts, securely log into the system, and manage personal information, learning preferences, and artistic goals. The module ensures that each user receives a personalized learning experience while maintaining the security and integrity of user data through appropriate authentication and authorization mechanisms.

4.4.5 Testing Strategy

Unlike conventional software testing, AtelierAI requires evaluation of both software components and Artificial Intelligence functionalities. Therefore, multiple testing strategies will be employed to ensure the reliability, usability, and effectiveness of the platform.

- i. **Functional Testing:** Functional testing will be conducted to verify that all system modules operate according to the specified requirements. This includes testing user authentication, roadmap generation, AI Art Mentor interactions, educational resource retrieval, exercise recommendations, and progress tracking functionalities.
- ii. **Retrieval Testing:** The Retrieval-Augmented Generation (RAG) pipeline will be evaluated to ensure that relevant educational resources are successfully

retrieved from the knowledge base. Testing will focus on retrieval accuracy, semantic search effectiveness, and the relevance of retrieved learning materials to user queries

- iii. **Response Quality Testing:** The AI Art Mentor will be evaluated by analyzing the quality, accuracy, and educational relevance of generated responses. Various drawing-related questions will be used to assess whether the system provides context-aware guidance grounded in retrieved educational resources.
- iv. **Usability Testing:** Usability testing will be performed with potential users to evaluate the ease of navigation, interface design, learning workflow, and overall user experience. Feedback collected during testing will be used to improve system usability and learner satisfaction.
- v. **Performance Testing:** Performance testing will be conducted to measure system responsiveness, API response time, database performance, and AI query processing speed. The objective is to ensure that the platform remains stable and responsive during normal operation.

5. Time Schedule

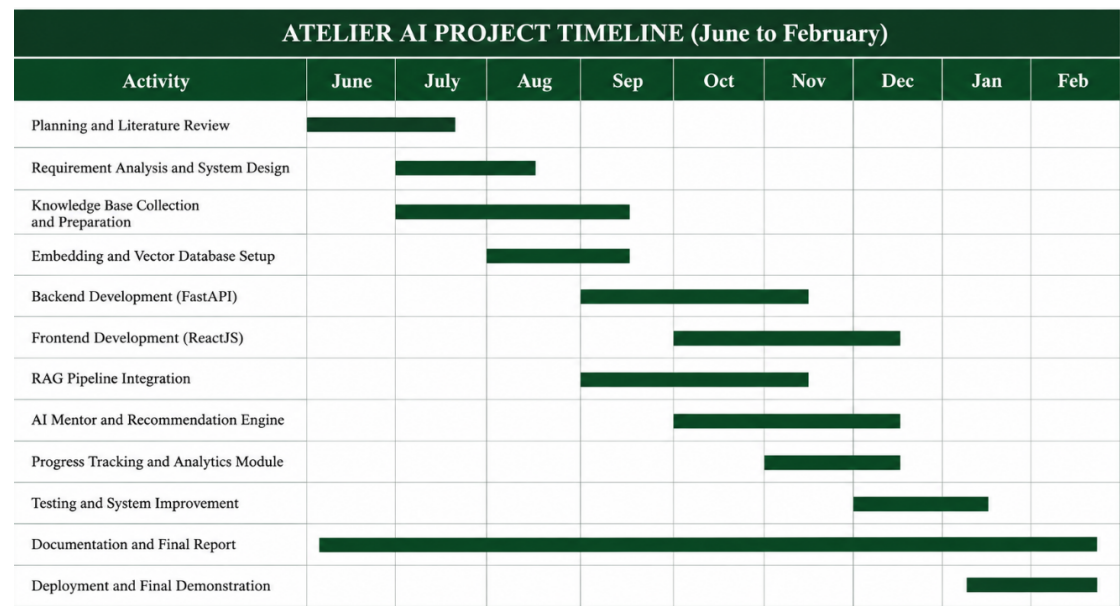


Figure 5.1: Proposed Gantt Chart

6. Expected Output

The expected output of AtelierAI is a fully functional web-based intelligent learning platform that assists aspiring artists through personalized educational guidance, Retrieval-Augmented Generation (RAG), and learning analytics. The platform is designed to support self-directed learning by providing structured roadmaps, educational resources, and AI-powered mentoring. The deliverables are categorized into functional outputs and academic contributions.

6.1 Functional Deliverables

- i. **AI Art Mentor Interface:** A web-based conversational interface that allows learners to ask drawing-related questions and receive context-aware educational guidance through Retrieval-Augmented Generation and Large Language Model integration.
- ii. **Personalized Learning Roadmap:** A structured learning pathway generated according to the learner's skill level, artistic interests, and learning objectives, providing systematic progression through drawing concepts and exercises.
- iii. **Educational Resource Management System:** A curated educational knowledge base containing drawing tutorials, perspective guides, anatomy references, and practice materials that support personalized learning and intelligent content retrieval.
- iv. **Exercise Recommendation and Progress Tracking System:** A learning analytics dashboard that monitors learner activities, tracks roadmap completion, visualizes progress, and recommends appropriate exercises and learning resources based on individual performance.

6.2 Technical Contributions

- i. **RAG Implementation:** Integration of Retrieval-Augmented Generation for context-aware educational assistance.

- ii. **Personalized Learning Framework:** Development of customized learning roadmaps based on learner profiles and goals.
- iii. **Intelligent Recommendation System:** Recommendation of exercises and learning resources based on learner progress.
- iv. **AI-Powered Educational Platform:** Development of a web-based platform integrating AI, semantic search, and learning analytics for art education.

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